

Homework 4

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Section 003
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Collaboration Statement

I worked on this homework independently. No outside collaborators or resources were used, except for the textbook *Elementary Analysis: The Theory of Calculus* by Ross and my own class notes.

Problem 1 (Ross Chapter 1, 4.7)

Let S and T be nonempty bounded subsets of $\mathbb{R}$.

(a) If $S \subseteq T$, then

$$\inf T \leq \inf S \leq \sup S \leq \sup T.$$

Proof. Assume $S \subseteq T$. We verify the three inequalities separately.

- i) $\sup S \leq \sup T$. Every element of S lies in T , so any upper bound of T is an upper bound of S . In particular, $\sup T$ is an upper bound of S . By minimality of the least upper bound, $\sup S \leq \sup T$.
- ii) $\inf T \leq \inf S$. Again $S \subseteq T$, hence any lower bound of T is a lower bound of S . In particular, $\inf T$ is a lower bound of S . By maximality of the greatest lower bound, $\inf T \leq \inf S$.
- iii) $\inf S \leq \sup S$. Because S is nonempty and bounded, there exists $s \in S$ and every $s \in S$ satisfies $\inf S \leq s \leq \sup S$. Hence $\inf S \leq \sup S$.

Putting the three statements together yields $\inf T \leq \inf S \leq \sup S \leq \sup T$. □

(b) Without assuming $S \subseteq T$,

$$\sup(S \cup T) = \max\{\sup S, \sup T\}.$$

Proof. Let $\alpha := \sup S$ and $\beta := \sup T$, and set $M := \max\{\alpha, \beta\}$.

Step 1: M is an upper bound for $S \cup T$. If $x \in S \cup T$, then either $x \in S$ or $x \in T$. In the first case $x \leq \alpha \leq M$; in the second $x \leq \beta \leq M$. Hence $x \leq M$ for all $x \in S \cup T$.

Step 2: M is the least such upper bound. Let u be any upper bound for $S \cup T$. Then u is an upper bound for S and for T , so $\alpha = \sup S \leq u$ and $\beta = \sup T \leq u$. Consequently $M = \max\{\alpha, \beta\} \leq u$.

By Steps 1–2, M is an upper bound for $S \cup T$ that is no larger than any other upper bound; by definition, $M = \sup(S \cup T)$. $\square$

Problem 2 (Ross Chapter 1, 4.8)

Let S and T be nonempty subsets of $\mathbb{R}$ with the property

$$(*) \quad s \leq t \quad \text{for all } s \in S \text{ and } t \in T.$$

(a) **Observation.** S is bounded above and T is bounded below.

Reason. Pick any $t_0 \in T$ (possible because $T \neq \emptyset$). By $(*)$, for every $s \in S$ we have $s \leq t_0$, so t_0 is an upper bound of S and hence S is bounded above.

Likewise, pick any $s_0 \in S$. Then for every $t \in T$ we have $s_0 \leq t$, so s_0 is a lower bound of T and hence T is bounded below. $\square$

(b) $\sup S \leq \inf T$.

Proof. By part (a), $\sup S$ and $\inf T$ exist (since $S \neq \emptyset$ is bounded above and $T \neq \emptyset$ is bounded below). Fix $t \in T$. Property $(*)$ implies $s \leq t$ for all $s \in S$, i.e., t is an upper bound for S . Hence $\sup S \leq t$ for every $t \in T$. Therefore $\sup S$ is a lower bound for T , and by maximality of $\inf T$, $\sup S \leq \inf T$. $\square$

(c) **Example with $S \cap T \neq \emptyset$.**

Take

$$S = [0, 1], \quad T = [1, 2].$$

Then for any $s \in S$ we have $s \leq 1 \leq t$ for every $t \in T$, so $(*)$ holds. Moreover, $S \cap T = \{1\} \neq \emptyset$.

(d) **Example with $\sup S = \inf T$ and $S \cap T = \emptyset$.**

Take

$$S = (0, 1), \quad T = [1, 2] \quad (\text{or } T = (1, 2]).$$

For any $s \in (0, 1)$ and any $t \in T$ we have $s < 1 \leq t$, so $(*)$ holds. Here $\sup S = 1 = \inf T$, while $S \cap T = \emptyset$ because $1 \notin S$ (and if $T = (1, 2]$ then $1 \notin T$ either).

Problem 3 (Ross Chapter 1, 4.12)

Let $\mathbb{I}$ be the set of real numbers that are not rational; elements of $\mathbb{I}$ are called *irrational numbers*. Prove: if $a < b$, then there exists $x \in \mathbb{I}$ such that $a < x < b$.

Lemma 1 (A rational in any open interval). *If $u < v$ are real numbers, then there exists $r \in \mathbb{Q}$ with $u < r < v$.*

Proof. By the Archimedean property, choose $n \in \mathbb{N}$ with $n(v - u) > 1$. There is an integer m with $nu < m < nv$ (e.g., take $m = \lfloor nv \rfloor$; then $m < nv$ and $m > nv - 1 > nu$). Set $r = \frac{m}{n} \in \mathbb{Q}$. Then

$$u < \frac{nu}{n} < \frac{m}{n} < \frac{nv}{n} < v,$$

so $u < r < v$. □

Lemma 2. *If $r \in \mathbb{Q}$, then $r + \sqrt{2} \in \mathbb{I}$.*

Proof. Suppose, toward a contradiction, that $r + \sqrt{2} \in \mathbb{Q}$. Since $\mathbb{Q}$ is closed under subtraction (is a field where the difference of rationals is rational?), $(r + \sqrt{2}) - r = \sqrt{2}$ would be rational, which contradicts the well-known fact that $\sqrt{2}$ is irrational. Hence $r + \sqrt{2}$ is irrational; i.e., $r + \sqrt{2} \in \mathbb{I}$. □

For any $a < b$ in $\mathbb{R}$, there exists $x \in \mathbb{I}$ with $a < x < b$.

Proof. Apply Lemma 1 to the interval $(a - \sqrt{2}, b - \sqrt{2})$ to obtain a rational $r \in \mathbb{Q}$ such that

$$a - \sqrt{2} < r < b - \sqrt{2}.$$

Let $x := r + \sqrt{2}$. Adding $\sqrt{2}$ to the inequalities gives

$$a < x < b.$$

By Lemma 2, $x = r + \sqrt{2}$ is irrational; hence $x \in \mathbb{I}$, as required. □

Problem 4 (Ross Chapter 1, 4.14(a))

4.14 Let A and B be nonempty bounded subsets of $\mathbb{R}$, and let $A + B$ be the set of all sums $a + b$ where $a \in A$ and $b \in B$.

- (a) Prove $\sup(A + B) = \sup A + \sup B$. *Hint:* To show $\sup A + \sup B \leq \sup(A + B)$, show that for each $b \in B$, $\sup(A + B) - b$ is an upper bound for A , hence $\sup A \leq \sup(A + B) - b$. Then show $\sup(A + B) - \sup A$ is an upper bound for B .

Proof. Since A and B are nonempty and bounded, $\sup A$ and $\sup B$ exist. Set $\alpha := \sup A$ and $\beta := \sup B$.

We prove the two inequalities separately.

- (i) $\sup(A + B) \leq \alpha + \beta$. For any $a \in A$ and $b \in B$ we have $a \leq \alpha$ and $b \leq \beta$, hence

$$a + b \leq \alpha + \beta.$$

Thus $\alpha + \beta$ is an upper bound for $A + B$, and by minimality of the least upper bound,

$$\sup(A + B) \leq \alpha + \beta.$$

- (ii) $\sup(A + B) \geq \alpha + \beta$. Fix $b \in B$. For every $a \in A$ we have $a + b \leq \sup(A + B)$, so

$$a \leq \sup(A + B) - b \quad (a \in A).$$

Hence $\sup(A + B) - b$ is an upper bound for A , and therefore

$$\alpha = \sup A \leq \sup(A + B) - b \quad \text{for every } b \in B,$$

which implies $\alpha + b \leq \sup(A + B)$ for every $b \in B$. Taking suprema over b gives

$$\sup_{b \in B} (\alpha + b) \leq \sup(A + B).$$

Since adding a constant preserves order, $\sup_{b \in B} (\alpha + b) = \alpha + \sup B = \alpha + \beta$. Hence

$$\alpha + \beta \leq \sup(A + B).$$

Combining (i) and (ii) yields $\sup(A + B) = \alpha + \beta = \sup A + \sup B$. □

Problem 5 (Ross Chapter 1, 4.15)

Let $a, b \in \mathbb{R}$. Show: if $a \leq b + \frac{1}{n}$ for all $n \in \mathbb{N}$, then $a \leq b$. (Compare Exercise 3.8.)

Proof. We argue by contradiction. Suppose $a > b$. Set $\varepsilon := a - b > 0$. By the Archimedean property, there exists $n \in \mathbb{N}$ with

$$\frac{1}{n} < \varepsilon = a - b.$$

Then

$$b + \frac{1}{n} < b + (a - b) = a,$$

which contradicts the hypothesis $a \leq b + \frac{1}{n}$ (true for every $n \in \mathbb{N}$). Therefore our assumption $a > b$ is impossible, and we conclude $a \leq b$. $\square$

Problem 6 (Ross Chapter 1, 4.16)

Show

$$\sup\{r \in \mathbb{Q} : r < a\} = a \quad \text{for each } a \in \mathbb{R}.$$

Proof. Let

$$S := \{r \in \mathbb{Q} : r < a\}.$$

Step 1. *S is nonempty and bounded above.* By the density of $\mathbb{Q}$ in $\mathbb{R}$, for any $u < v$ there exists $q \in \mathbb{Q}$ with $u < q < v$. Taking $u = a - 1$, $v = a$, we get $q < a$, so $S \neq \emptyset$. Since every $r \in S$ satisfies $r < a$, the real number a is an upper bound for S . Hence $\sup S$ exists and $\sup S \leq a$.

Step 2. *a is the least upper bound.* Let $\varepsilon > 0$. By the density of $\mathbb{Q}$, there exists $q \in \mathbb{Q}$ with

$$a - \varepsilon < q < a.$$

Then $q \in S$ and $q > a - \varepsilon$. Thus $a - \varepsilon$ is not an upper bound of S . Therefore $\sup S \geq a - \varepsilon$. Since this is true for every $\varepsilon > 0$, we conclude $\sup S \geq a$.

Combining Step 1 and Step 2 gives

$$\sup S = a.$$

$\square$